

Table 3: Explanation results for node classification. Standard deviation estimated from 10 independent runs, with sampling and training repeated for each run. Best result in **bold**, second best is underlined.

	Models	GCN		GAT		GIN		GraphSAGE	
		IoU ($\uparrow$)	Pr ($\uparrow$)	IoU ($\uparrow$)	Pr ($\uparrow$)	IoU ($\uparrow$)	Pr ($\uparrow$)	IoU ($\uparrow$)	Pr ($\uparrow$)
<i>BA-shapes</i>	Random Explainer	0.0029 $\pm$.01	0.0052 $\pm$.01	0.0037 $\pm$.00	0.0062 $\pm$.01	0.0032 $\pm$.00	0.0057 $\pm$.01	0.0040 $\pm$.00	0.0071 $\pm$.00
	IntegratedGradient	0.6590 $\pm$.00	0.7714 $\pm$.00	0.7902 $\pm$.00	0.8619 $\pm$.00	0.5961 $\pm$.00	0.7238 $\pm$.00	0.8424 $\pm$.00	0.9000 $\pm$.00
	InputXGradient	0.7021 $\pm$.00	0.8000 $\pm$.00	<u>0.8299$\pm$.00</u>	<u>0.8857$\pm$.00</u>	0.5848 $\pm$.00	0.7143 $\pm$.00	<u>0.8424$\pm$.00</u>	<u>0.9000$\pm$.00</u>
	PGExplainer	<u>0.8163$\pm$.05</u>	<u>0.8710$\pm$.10</u>	0.5731 $\pm$.05	0.6395 $\pm$.06	0.8566$\pm$.03	0.8929$\pm$.02	<u>0.6378$\pm$.14</u>	<u>0.7210$\pm$.12</u>
	GNNExplainer	0.5683 $\pm$.02	0.7019 $\pm$.01	0.2922 $\pm$.01	0.3967 $\pm$.01	0.5372 $\pm$.03	0.6614 $\pm$.02	0.4487 $\pm$.03	0.5976 $\pm$.02
	SubgraphX	0.5363 $\pm$.01	0.6729 $\pm$.01	0.5935 $\pm$.03	0.7010 $\pm$.02	0.7369 $\pm$.02	0.8181 $\pm$.02	0.6036 $\pm$.02	0.7338 $\pm$.01
	CIExplainer (Ours)	0.8909$\pm$.02	0.9314$\pm$.01	0.8448$\pm$.01	0.8738$\pm$.01	<u>0.7915$\pm$.02</u>	<u>0.8667$\pm$.01</u>	0.9294$\pm$.01	0.9576$\pm$.04
<i>Tree-grid</i>	Random Explainer	0.0036 $\pm$.00	0.0067 $\pm$.00	0.0053 $\pm$.00	0.0099 $\pm$.01	0.0040 $\pm$.00	0.0075 $\pm$.00	0.0024 $\pm$.00	0.0046 $\pm$.00
	IntegratedGradient	0.8868 $\pm$.00	0.9361 $\pm$.00	<u>0.8395$\pm$.00</u>	<u>0.8976$\pm$.00</u>	0.8741$\pm$.00	0.9285$\pm$.00	<u>0.8868$\pm$.00</u>	<u>0.9361$\pm$.00</u>
	InputXGradient	<u>0.8887$\pm$.00</u>	<u>0.9376$\pm$.00</u>	0.8381 $\pm$.00	0.8954 $\pm$.00	0.0182 $\pm$.00	0.0228 $\pm$.00	<u>0.8890$\pm$.00</u>	<u>0.9376$\pm$.00</u>
	PGExplainer	0.0971 $\pm$.17	0.1335 $\pm$.20	0.0333 $\pm$.00	0.0580 $\pm$.02	0.8701 $\pm$.02	0.9253 $\pm$.01	0.1960 $\pm$.06	0.1960 $\pm$.07
	GNNExplainer	0.7754 $\pm$.01	0.8658 $\pm$.01	0.7413 $\pm$.01	0.8382 $\pm$.00	0.0182 $\pm$.00	0.0228 $\pm$.00	0.7449 $\pm$.01	0.8397 $\pm$.00
	SubgraphX	0.2985 $\pm$.01	0.4464 $\pm$.01	0.3042 $\pm$.01	0.4534 $\pm$.02	0.3118 $\pm$.01	0.4632 $\pm$.02	0.3124 $\pm$.02	0.4647 $\pm$.02
	CIExplainer (Ours)	0.9008$\pm$.01	0.9381$\pm$.01	0.8439$\pm$.00	0.9075$\pm$.01	<u>0.8407$\pm$.00</u>	<u>0.8988$\pm$.00</u>	0.9325$\pm$.02	0.9595$\pm$.01

Table 5: Explanation results for graph classification. Standard deviation obtained from 10 independents explanations. Best result in **bold**, second best is underlined

	Models	GCN		GAT		GIN		GraphSAGE	
		IoU ($\uparrow$)	Pr ($\uparrow$)	IoU ($\uparrow$)	Pr ($\uparrow$)	IoU ($\uparrow$)	Pr ($\uparrow$)	IoU ($\uparrow$)	Pr ($\uparrow$)
<i>BA-2motif</i>	Random Explainer	0.1224 $\pm$.01	0.2020 $\pm$.01	0.1234 $\pm$.01	0.2036 $\pm$.02	0.1235 $\pm$.01	0.2048 $\pm$.02	0.1171 $\pm$.01	0.1942 $\pm$.02
	IntegratedGradient	0.0022 $\pm$.00	0.0040 $\pm$.00	0.0847 $\pm$.00	0.1520 $\pm$.00	0.0000 $\pm$.00	0.0000 $\pm$.00	0.1652 $\pm$.00	0.2700 $\pm$.00
	InputXGradients	0.0111 $\pm$.00	0.0200 $\pm$.00	0.1225 $\pm$.00	0.2120 $\pm$.00	0.0078 $\pm$.00	0.0140 $\pm$.00	0.2815 $\pm$.00	0.4280 $\pm$.00
	PGExplainer	<u>0.6369$\pm$.08</u>	<u>0.7216$\pm$.05</u>	0.7696$\pm$.08	0.8272$\pm$.07	<u>0.3859$\pm$.02</u>	<u>0.5210$\pm$.02</u>	0.8823$\pm$.08	0.9240$\pm$.06
	GNNExplainer	0.1144 $\pm$.01	0.1896 $\pm$.02	0.0738 $\pm$.01	0.1264 $\pm$.02	0.1056 $\pm$.01	0.1772 $\pm$.02	0.1129 $\pm$.01	0.1880 $\pm$.01
	SubgraphX	0.3015 $\pm$.02	0.3554 $\pm$.03	<u>0.3108$\pm$.04</u>	<u>0.3676$\pm$.03</u>	0.3301 $\pm$.04	0.3864 $\pm$.04	0.3192 $\pm$.03	0.3726 $\pm$.03
	CIExplainer (Ours)	0.6745$\pm$.02	0.7842$\pm$.01	0.1349 $\pm$.00	0.2292 $\pm$.01	0.3921$\pm$.01	0.5498$\pm$.01	<u>0.7503$\pm$.01</u>	<u>0.8336$\pm$.01</u>
<i>MUTAG</i>	Random Explainer	0.1107 $\pm$.01	0.1747 $\pm$.02	0.1084 $\pm$.00	0.1719 $\pm$.01	0.1105 $\pm$.01	0.1751 $\pm$.02	0.1090 $\pm$.01	0.1723 $\pm$.02
	IntegratedGradient	<u>0.4459$\pm$.00</u>	<u>0.5580$\pm$.00</u>	0.1202 $\pm$.00	0.1985 $\pm$.00	<u>0.1283$\pm$.00</u>	<u>0.2040$\pm$.00</u>	0.3025$\pm$.00	0.4246$\pm$.00
	InputXGradients	0.0827 $\pm$.00	0.1100 $\pm$.00	0.1208 $\pm$.00	0.1748 $\pm$.00	0.1281 $\pm$.00	0.1982 $\pm$.00	0.1097 $\pm$.00	0.1683 $\pm$.00
	PGExplainer	0.0780 $\pm$.04	0.1200 $\pm$.06	0.1132 $\pm$.03	0.1658 $\pm$.00	0.1000 $\pm$.01	0.1549 $\pm$.05	<u>0.1650$\pm$.09</u>	<u>0.2343$\pm$.13</u>
	GNNExplainer	0.6251$\pm$.02	0.7225$\pm$.01	0.1659$\pm$.01	0.2467$\pm$.02	0.0611 $\pm$.01	0.0950 $\pm$.01	0.1872 $\pm$.01	0.2871 $\pm$.02
	SubgraphX	0.0808 $\pm$.01	0.1159 $\pm$.01	0.0877 $\pm$.01	0.1259 $\pm$.01	0.0861 $\pm$.01	0.1235 $\pm$.01	0.0863 $\pm$.01	0.1240 $\pm$.02
	CIExplainer (Ours)	0.1160 $\pm$.01	0.1844 $\pm$.02	<u>0.1314$\pm$.01</u>	<u>0.2141$\pm$.02</u>	0.1751$\pm$.01	0.2580$\pm$.01	0.1213 $\pm$.01	0.1890 $\pm$.01